Build Custom AI

Course and Project Guide | RiskLens

Competency-based, fully self-paced.

Build Custom AI teaches you to adapt pretrained models into a measurable AI system. Across five modules, you learn model foundations and vision-language models, retrieval-augmented generation, QLoRA fine-tuning, model optimization, and end-to-end evaluation through guided Google Colab practice.

This guide explains what you build, how the five modules fit together and what evidence to keep. The RiskLens Grading Rubric holds the assessment standards. Published course policy governs attempts, timing and academic procedures.

The course at a glance

Item	What to expect
Course project	RiskLens: one cumulative product-risk pipeline.
Modules and competencies	Five modules and five competencies, with 24 Met / Not Yet criteria. These groups do not establish credit units.
Learning format	40 short videos, 6-8 minutes each; readings and guided practice in Google Colab.
Starting knowledge	Working Python and notebook skills; basic familiarity with machine-learning evaluation.
Tools and runtime	Google Colab, Hugging Face, a GPU runtime, retrieval tools and saved model artifacts. The starter targets a 16 GB T4.
Assessment	Five milestone submissions of the same notebook, with saved outputs and accessible artifacts.
Outcome	A reproducible pipeline and evidence explaining its quality, limitations and deployment choice.

What you will build

One system developed across the course

RiskLens is a product-risk analysis pipeline. It inspects a product photo for visible defects, retrieves supporting reviews and specifications, and uses a fine-tuned Analyst to produce a structured root-cause and recommended-action report. You improve the inherited components, evaluate them on frozen data, compare compressed models, and assemble one reproducible system.

How the components work together

- A product image enters the VLM inspector, which returns a verdict and defect description. A good product follows the documented short-circuit path.

- For a defective product, the retriever finds relevant reviews and specifications. Retrieved source IDs remain attached so the report can be checked.

- The QLoRA Analyst receives the prose defect note and evidence in the same input format used for training. It produces the required root-cause report sections.

- Compressed candidates are compared with the trained reference. The final pipeline uses an accepted arm or a clearly disclosed, feasible fallback.

What is assessed

You must demonstrate both the system and your reasoning. Working components alone are insufficient: keep valid comparisons, frozen artifacts, failure analyses and the explanation of your choices. Attractive model outputs cannot replace a measured result.

What to keep as you work

- Prompts, representative raw outputs, data and model identifiers, configuration settings and dated notebook snapshots.

- Development/test membership, gold questions, dataset card, adapter paths and the predeclared gate.

- Saved evaluation results and the metrics ledger. Export per-run comparisons separately because recording the same ledger key overwrites its previous value.

Your path through the modules

Complete the modules in order

Module 1 - Visual Defect Inspector

Competency: Build and evaluate a vision-language inspector that diagnoses and corrects the inherited baseline defects, returns reliable structured outputs, improves defect F1 on a frozen test set, and justifies prompting and image-resolution choices using measured quality and cost.

Submit: D1.1-D1.7. All C1 criteria are assessed at this milestone.

Module 2 - RAG Evidence Engine

Competency: Build and evaluate a retrieval-augmented evidence engine that finds relevant product information, supports its answers with auditable citations, refuses unsupported requests, and justifies retrieval and graph-based design choices using controlled measurements.

Submit: D2.1-D2.7. All C2 criteria are assessed at this milestone.

Module 3 - Fine-Tuned Analyst

Competency: Curate and document teacher-generated training data, train and reload a QLoRA Analyst, and evaluate it against the untuned base model on separated data using format, content and pairwise comparisons, with a controlled ablation and honest limitations.

Submit: D3.1-D3.7. All C3 criteria are assessed at this milestone.

Module 4 - Optimized Analyst

Competency: Explain parallelism and model-optimization options, compare compressed versions of the trained Analyst under a predeclared quality gate, and justify an accept, reject or fallback decision using measured quality, memory, latency, throughput and cost.

Submit: D4.1-D4.4. All C4 criteria are assessed at this milestone.

Module 5 - Integrated RiskLens and Final Delivery

Competency: Integrate the learner-built inspector, evidence engine and selected Analyst into a reproducible RiskLens pipeline, evaluate its behavior and failures across products, and explain its architecture, model choice and measurement limits using the submitted evidence.

Submit: D4.5-D4.7 plus final delivery. All C5 criteria are assessed at this milestone.

Working in the project notebook

Preserve the contracts and the evidence

Start with one copy

Save your own copy of capstone_starter.ipynb, set the team identifier and read Section 0. Keep the same shareable notebook link throughout the course. Complete the labeled scaffolds and add supporting cells below them. Preserve the GIVEN functions and the cross-stage contracts.

The repository still uses four stage labels and older weekly grading language. In this course, D1.1-D3.7 map to Modules 1-3; D4.1-D4.4 map to Module 4; D4.5-D4.7 and final delivery map to Module 5. The v2.0 rubric supersedes legacy points, deductions and bonuses.

Separate practice from assessment

Lesson notebooks teach the techniques and provide guided examples. The course project requires your own completed scaffolds, real teacher-model data, saved model artifacts and measured results. A template response or an integration stub is not evidence of the required component.

The Module 4 checkpoint

At Module 4, add a standalone checkpoint below your workspaces for the frozen-set format/content comparison, benchmarks and verdict. Save these outputs without calling analyze_product(). Run the starter's combined Stage 4 evaluation after integration in Module 5. Do not edit GIVEN functions or use stubs to satisfy the checkpoint.

Run and save without retraining

At each milestone, execute the completed work through that module and save its outputs; later incomplete scaffolds do not make an earlier milestone incomplete. For final delivery, use a train-or-load path and a clean top-to-bottom run. Keep model artifacts accessible after the temporary runtime closes.

The starter keeps the VLM resident and asks for an Analyst no larger than 1.5B parameters with a 4-bit serving arm. Load comparison arms sequentially when needed and report actual memory use. A fallback must fit the documented runtime; a hypothetical model is not a runnable fallback.

How mastery and revision work

Read the criteria before starting a module

Met and Not Yet

Each criterion is Met or Not Yet. A competency is Mastered only when every criterion in that group is Met. The course project is Mastered when all five competencies are Mastered. There are no weighted modules, points, averages, deductions or bonus credit. Model evaluation scores remain technical evidence, not course grades.

Review order

Review the MUST PASS criterion first for each module. If it is Not Yet, return that submission with targeted feedback before reviewing the remaining criteria. Every other criterion is also required; MUST PASS determines review order, not extra weight.

Feedback and resubmission

Feedback identifies each Not Yet criterion, the missing or incorrect evidence and the required revision. Resubmit a short revision plan and the revised evidence. Reassess Not Yet criteria and any previously Met evidence affected by the changes. Attempt limits, waiting periods, enrolment rules and formal grade codes follow the published course policy.

Incomplete or unverifiable evidence

Return an inaccessible or incomplete submission for correction before assessment. Missing required outputs, a broken run/load path or exposed credentials must be resolved. Refer fabricated measurements, reconstructed evidence or copied work for academic review under course policy; do not convert these into point deductions.

A negative experiment can still be useful

The inspector must beat baseline defect F1, and the retriever must beat baseline recall@4. The Analyst has no invented minimum win-rate. A correctly evaluated tuning or compression experiment may produce worse results; report them and justify the next decision. A failed gate cannot be presented as deployment approval.

No hidden-holdout threshold, timed task or recording protocol is assumed. The final notebook includes an evidence-based explanation under C5.4; separately issued exam instructions may map equivalent evidence to that criterion.

Final submission and useful habits

Make the evidence easy to review

Final delivery checklist

- The notebook link opens for the reviewer and the submitted version is identified.

- All required scaffolds are complete; final Run all loads saved weights and completes without hidden runtime state.

- Evaluation outputs, intermediate pipeline results and show_ledger() are visible. Earlier experiment results remain available.

- Model and data artifacts, environment details and run/load instructions are accessible.

- D4.5 product demonstrations, D4.7 batch results and D4.6 deployment note are present.

- The reflection is at most 300 words; the C5.4 walkthrough explains the submitted component, product trace, metric and limitation.

How to do well

Freeze before experimenting, preserve comparisons instead of overwriting them, and read the actual failures. State what a number measures and what it cannot prove. Keep your explanation close to your saved code and outputs. If a result fails the stated standard, revise the work and identify what changed.

Where to find the standard

- Build Custom AI Course Project v2: module goals, full competencies, videos, readings, deliverables and criteria.

- RiskLens Grading Rubric v2.0: all 24 criteria and assessment rules in one file.

- RiskLens Module 1-5 Project Rubrics: evidence checklist, full Met standards and common mistakes for each module.

- Published course policy: attempts, waiting periods, enrolment and academic procedures. Use the designated course support channel for clarification.

Project source: Building Custom AI project repository. Assessment mapping reviewed against commit c70b3de1145d.